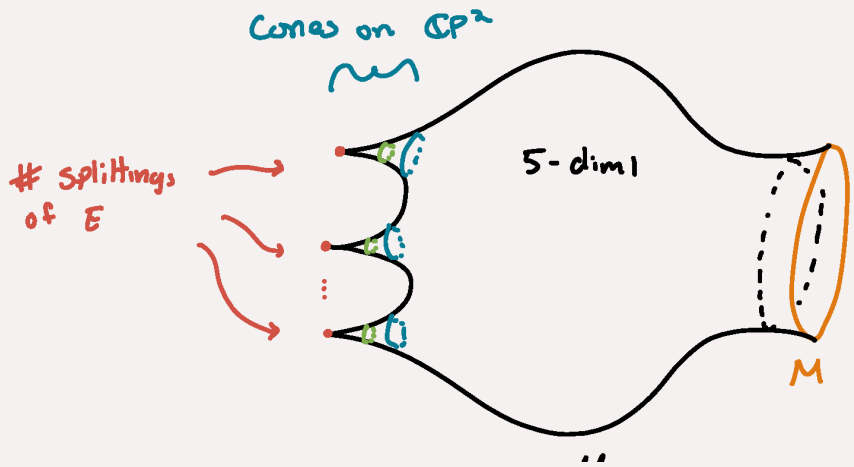
Lecture 8 // Sept 17 // Singularities in $\mathcal{M}$ & Holonomy

Last Time: Let (M^4, g) be positive definite and simply connected. Fix a $k = 1$ SU(2) bundle E over M. Consider $\mathcal{M} = \mathcal{A}/\mathcal{G}$ the SDYM moduli space. We know the following qualitative features of $\mathcal{M}$:

1. Away from a finite number of singular points, $\mathcal{M}$ is a smooth 5-dimensional oriented manifold.
2. The number of these points = the number of splittings of E.
3. Each point has a neighborhood homeomorphic to the cone on $\mathbb{CP}^2$; diffeomorphic away from the point.
4. $\mathcal{M}$ is not compact, but it admits a compactification $\overline{\mathcal{M}}$ such that $\partial\overline{\mathcal{M}} \cong M$.



After some linear algebra, these properties prove Donaldson's theorem: since σ is a bordism invariant,

$$\underbrace{\operatorname{rk} Q_M}_{\text{since definite}} = \sigma(M) = \#\text{ splittings of } E = \frac{1}{2}\#\{x \in H_2 \text{ with } x^2 = 1\}.$$

Today, we'll unpack these points: why is $\mathcal{M}$ singular, and why do neighborhoods look like the cone on $\mathbb{CP}^2$?

Singularities in $\mathcal{M}$

Recall that $\mathcal{M} = \{\text{SDYM conn.}\}/\mathcal{G}$. $\{\text{SDYM conn.}\} \subseteq \mathcal{A}$, which is a smooth manifold, on which $\mathcal{G}$ acts. However, $\mathcal{G}$ may have stabilizer subgroups. Thus the singularities arise from the failure of $\mathcal{G}$ to act freely on $\mathcal{A}$.

Remark 1
Establishing {SDYM conn.} is a smooth manifold is fairly standard: it is the zero locus of a map from an ∞-dimensional manifold to somewhere else; verify that 0 is a regular value, apply Sard-Smale.

Definition 2
For $\nabla \in \mathcal{A}$, $\mathrm{Stab}(\nabla) = \{g \in \mathcal{G} \mid g^*(\nabla) = \nabla\}$,

$$A \longrightarrow g^{-1}Ag + g^{-1}dg.$$

What can $\mathrm{Stab}(\nabla)$ be? First note that for every ∇, $\pm\mathrm{Id} \in \mathrm{Stab}(\nabla)$. Since $\pm\mathrm{Id}$ is sent to itself under conjugation, $\pm\mathrm{Id}$ is a well-defined section of $\mathrm{Aut}(E)$. These are not interesting gauge transformations, so it's sometimes more convenient to take $\mathcal{G}/\pm\mathrm{Id} \curvearrowright \mathcal{A}$ instead.

What else is possible? Let's examine an example. Connections with nontrivial stabilizers are related to the number of splittings. For a line bundle L, consider the constant gauge transformation given by multiplication by a fixed element of S^1, $e^{i\theta} \in S^1$. For $E = L \oplus \bar{L}$, consider now

$$g = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}.$$

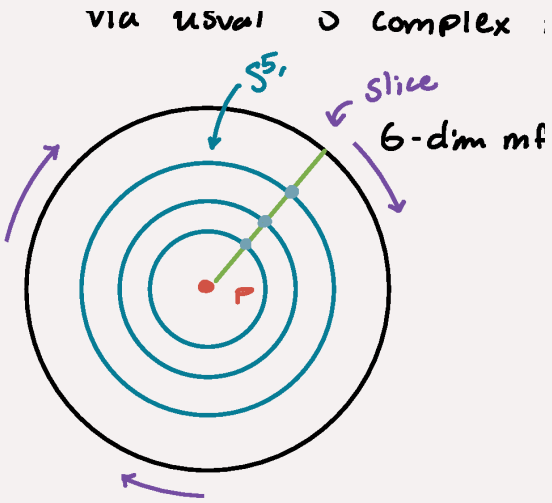
One can check this is a well-defined gauge transformation on E. If ∇ is split, e.g., $A = \begin{pmatrix} ia & 0 \\ 0 & -ia \end{pmatrix}$, easy to check that $g^*(\nabla) = \nabla$. So if E is split and ∇ is a split connection, considered as a subgroup of $\mathrm{SU}(2)$, $\mathrm{Stab}(\nabla)$ is at least as big as S^1.

Remark 3
In this observation, we play fast and loose with the difference between a local definition of a gauge transformation and its definition globally. The correct statement is given in the following lemma, and the tension between global & local will be resolved as we introduce the tools to prove it.

Lemma 4
For an $\mathrm{SU}(2)$ bundle with $k \neq 0$, a connection has stabilizer bigger than $\pm\mathrm{Id}$ iff it is split, and $\mathrm{Stab}(\nabla) \cong S^1$.

The proof isn't hard, but requires more technical language. But this lemma gives why the singularities of $\mathcal{M}$ are what they are. We showed previously that each splitting of E supports a unique (up to $\mathcal{G}$) split SDYM connection. Each singular point is the unique SDYM connection for a splitting. Thus the number of singular points is the number of splittings. This gives (2).

We now ask why local neighborhoods are cones on $\mathbb{CP}^2$. Near a singular point, $\mathcal{G}/\pm I$ acts freely. The approximate idea is that we first quotient by $\pm I$ to get a 6-dimensional manifold with a $S^1/\pm I$ action which acts freely except at one point. The local model for this is $S^1 \curvearrowright \mathbb{R}^6 \cong \mathbb{C}^3$ via usual S^1 complex multiplication. $\mathbb{C}^3/S^1 \cong$ cone on $\mathbb{CP}^2$ with one singular point, since on each S^5 slice, we have $S^5/S^1 \cong \mathbb{CP}^2$ (via definition $\mathbb{CP}^n := S^{2n+1}/S^1$).



Note 5
The $\pm I \subset \mathcal{G}$ and S^1 actions are more entertwined in $\mathcal{G}$ than this; we can't actually quotient by gauge in this "partial" way. But it gives an approximate idea.

Question 6. *Why is $\mathcal{M}$ 5-dimensional?*

This is where we use that $k = 1$. The dimension is from the Atiyah-Singer index formula applied to a Fredholm operator whose zero locus are the SDYM connections. When $k > 1$, we get more generally that the singular neighborhood is $\cong$ cone on $\mathbb{CP}^n$.

We now must introduce more differential geometry.

Interlude: Parallel Transport

Let E be a vector bundle over M.

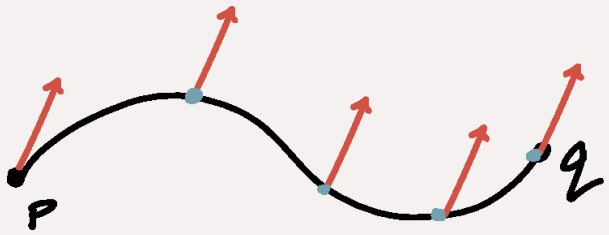
Definition 7
Let ∇ be a connection on E, $\gamma : [0,1] \to M$ a path. A section s defined along γ is parallel transported by ∇ along γ if $\nabla_{\gamma'(t)} s = 0$ for all t.

Lemma 8
Let $\gamma : [0,1] \to M$ be a path from p to q. If we fix any $v \in E_p$, then there exists a unique section s defined along γ such that s extends v and s is parallel transported by ∇ along γ.

Proof idea. Using the existence of solutions to ODEs, we can construct s and show that $(d + A)s = 0$. □

To better see the point of parallel transport, we can observe the effect of parallel transport on a frame of E_p, i.e., a basis for E_p, and parallel transport that basis along γ. Parallel transport preserves

orthonormality, so this gives you a particular trivialization of E along γ for which the basis elements are covariantly constant: A vanishes in the path direction γ'. In essence, the covariant derivative specifies a "preferred frame" along each path, which we can find by choosing a frame at one end and parallel transporting it along the path.



What else can parallel transport do? For each path γ from p to q, we can build a linear map from E_p to E_q which sends $v \in E_p$ to the endpoint of the parallel transport of v along γ. Call this map $\mathrm{hol}_{p,q}(\gamma) : E_p \to E_q$. Since $\mathrm{hol}_{p,q}(\gamma) \circ \mathrm{hol}_{q,p}(\bar{\gamma}) = \mathrm{id}$, this map is an isomorphism.

Now, take $q = p$, so γ is a based loop in M. Then $\mathrm{hol}_p = \mathrm{hol}_{p,p}$ gives a map $E_p \to E_p$, so defines an element $\mathrm{hol}_p(\gamma) \in \mathrm{GL}(E_p)$.

Definition 9
Holonomy is a map $\mathrm{hol}_p : \{\text{loops based at } p\} \to \mathrm{GL}(E_p)$.

Properties of Holonomy:

1. If E is a G-bundle and ∇ is a G connection, then $\mathrm{hol}_p(\gamma)$ is $G \subseteq \mathrm{GL}(E_p)$ valued.
2. If $F_\nabla = 0$, the holonomy only depends on the based smooth homotopy class of γ. I.e., hol_p descends to a map
$$\mathrm{hol}_p : \pi_1(M) \to \mathrm{GL}(E_p) \quad (\text{or } G \text{ for } G).$$
For $G = \mathrm{SU}(2)$, $\mathrm{hol}_p : \pi_1(M) \to \mathrm{SU}(2)$ is a $\mathrm{SU}(2)$ representation of $\pi_1(M)$.
3. If g is a gauge transformation,
$$\mathrm{hol}_{p,q}^{g^*(\nabla)}(\gamma) = g_q^{-1} \circ \mathrm{hol}_{p,q}^{\nabla}(\gamma) \circ g_p.$$

To illustrate holonomy, consider a trivial line bundle $L = M \times \mathbb{C}$ over M and a $U(1)$ connection on L given by $\nabla = d + ia$, $a \in \Omega^1(M; \mathbb{R})$. Thus a section here is just a $\mathbb{C}$-valued function. For a path γ, what does it mean for $\nabla_{\gamma'(t)} s = 0$? Given a based loop γ and a section defined along γ, what does it mean for s to be parallel transported along γ? Since we're working in a line bundle, a section is just a $\mathbb{C}$-valued function, and for $\nabla_{\gamma'(t)} s = 0$, the covariant derivative of this function must vanish along γ. Explicitly, we get

$$\nabla_{\gamma'(t)} s = 0, \qquad (ds)(\gamma'(t)) + ia(\gamma'(t))s = 0, \qquad \frac{d}{dt}(s \circ \gamma) + ia(\gamma'(t))(s \circ \gamma(t)) = 0.$$

But this is just a differential equation, which we can solve as usual:

$$\frac{d}{dt}(s \circ \gamma) = -ia(\gamma'(t))(s \circ \gamma(t)) \;\Rightarrow\; (s \circ \gamma) = \left(e^{-i\int_0^t a(\gamma'(h))\,dh}\right) s(\gamma(0)).$$

If we conflate s and $s \circ \gamma$, we get that $\text{hol}_{p,q} = e^{-i\int_\gamma a} \in S^1$.

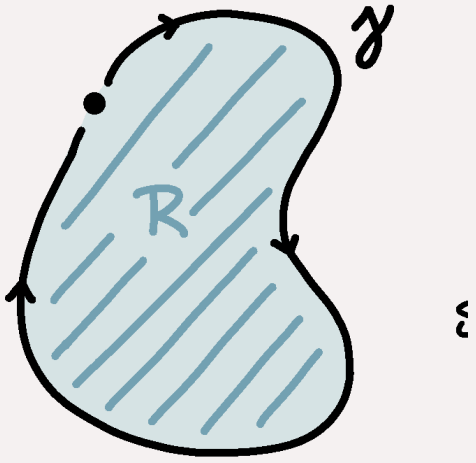
Intuitively, we're supposed to think that solving $(d + A)s = 0$ gives $s(t) = e^{-\int A} s(0)$, so if A is Lie algebra valued, $-\int A$ is Lie algebra valued, so $e^{-\int A}$ is Lie group valued. In reality, this is false – there's an issue with commutativity when $G \neq U(1)$.

However we see directly here that $\text{hol}(\gamma)$ need not be the identity: if $\int_\gamma a \neq 2\pi n$, $e^{-i\int_\gamma a} \neq 1$.

In fact we see a nice connection with property #3. Indeed, suppose that γ bounds a region R. Then Stokes' theorem says that

$$\int_\gamma a = \int_{\partial R} a = \int_R da,$$

where da is the curvature of ∇.



So the heuristic is that holonomy picks up the curvature of a connection.

This demonstrates, in the $U(1)$ case, assertion #3: if γ is varied through a region where $F_\nabla = 0$, then holonomy is not changed. However, note that this analysis applies to single coordinate charts; it is purely local. If the bundle is nontrivial, it is not true that $F_A = 0$ implies that $\text{hol}(\gamma) = \text{id}$. Furthermore, it makes no sense to treat the differential equation like it's globally defined, since a isn't. But it is true that holonomy is constant in its homotopy class.

Next time: We'll use parallel transport to discuss the lemma on the exact behavior of stabilizers of connections.